

Homework 7: Computed Torque Control of a 4-DOF Manipulator

Manipulator: 3R+1P (MLS Example 4.4). Link lengths: $l_1 = 0.9$ m, $l_2 = 0.7$ m, $l_0 = 0.3$ m. Masses: 35, 25, 18, 7 kg. $\beta = [0.9, 0.45, 0.4, 0.9]$. $\theta(0) = [0.3\pi, 0.2\pi, 0.35\pi, -0.03]^\top$.
Desired final values: $[2, 1, 0.5, 0.08]^\top$ (rad / m). **Gains:** $K_p = 2500I$, $K_v = 100I$ ($\omega_n = 50$ rad/s, $\zeta = 1$). **Saturation:** $|\tau| \leq [6000, 2000, 1000, 100]^\top$.

1 Part (a): Computed Torque Controller

Method. The computed torque law $\tau = M(\theta)\ddot{\theta}^* + C\dot{\theta} + N$ with $\ddot{\theta}^* = \ddot{\theta}_d + K_v\dot{e} + K_p e$ exactly cancels the nonlinear dynamics, yielding linear error dynamics $\ddot{e} + K_v\dot{e} + K_p e = 0$. The desired trajectory is a quintic rounded step starting at $t_0 = 1.0$ s with $\Delta t = 0.4$ s, scaling a unit step to the specified amplitudes. Dynamics are computed via screw theory (PoE) in MCNsim.

Results.

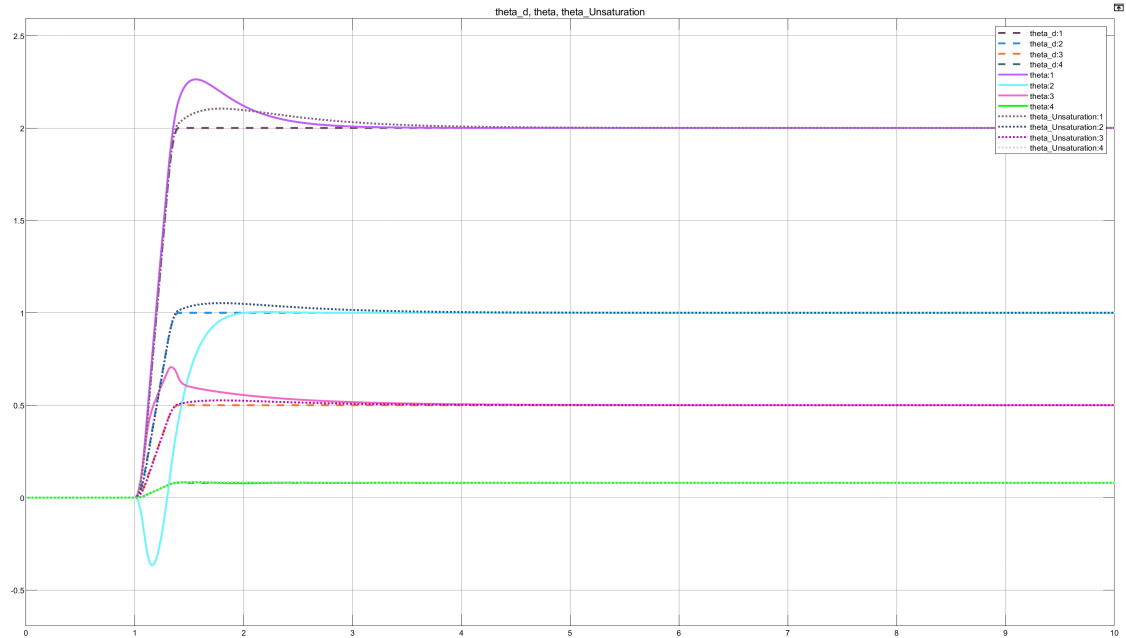


Figure 1: Nominal computed torque tracking. **Dashed:** θ_d (desired). **Solid:** θ (actual, with saturation). **Dash-dot:** θ_{unsat} (without saturation).



Figure 2: Control torques $\tau(t)$ with saturation limits $[6000, 2000, 1000, 100]^\top$ shown.

Observations:

- All joints converge to desired values: $\theta_1 \rightarrow 2$, $\theta_2 \rightarrow 1$, $\theta_3 \rightarrow 0.5$ rad, $d_4 \rightarrow 0.08$ m.
- Errors converge exponentially ($\tau \approx 0.02$ s) to zero — exact cancellation yields perfect asymptotic tracking.
- Saturation has a visible effect: θ (with saturation) deviates from θ_{unsat} (without saturation), most notably on joint 2 where a temporary dip occurs — indicating that the torque limits $[6000, 2000, 1000, 100]^\top$ are occasionally hit during the transient (see also §4).
- Torque peaks during step transition $t \in [1, 1.4]$ s; joint 1 requires the largest torque.

2 Part (b): Robustness to Parameter Mismatch

Method. We perturb the plant spatial inertias by $\delta \in \{-10\%, -5\%, +5\%, +10\%\}$ while keeping the controller at nominal values — simulating aging-induced parameter drift. Saturation limits $[6000, 2000, 1000, 100]^\top$ are active. Figure 3 shows the case $\delta = +10\%$ (plant heavier by 10%).

Results.

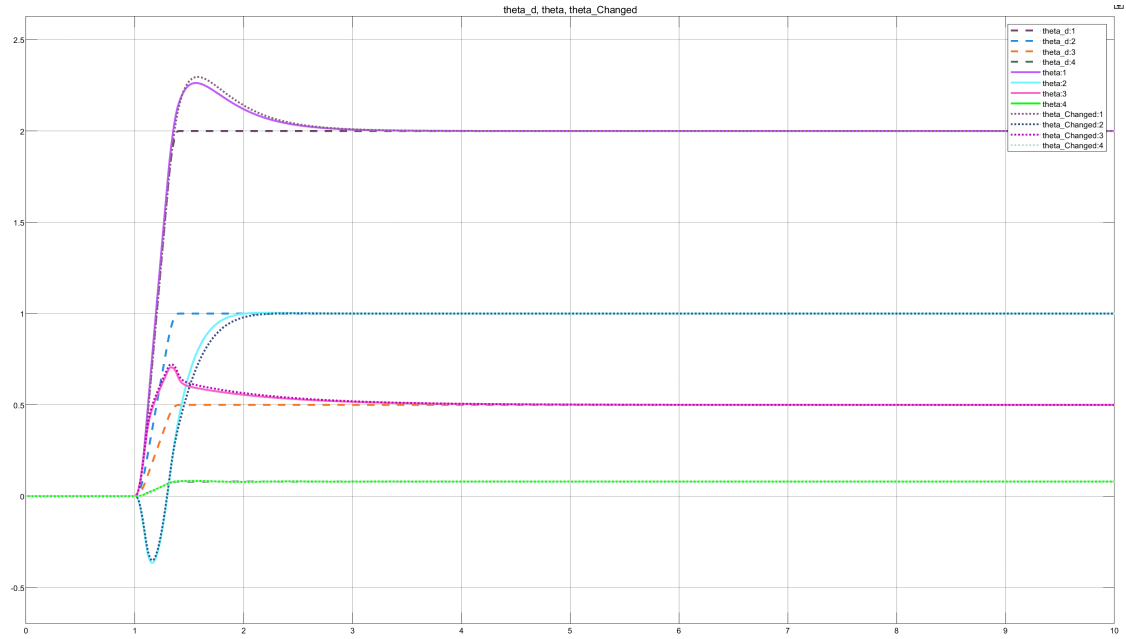


Figure 3: Robustness under parameter perturbation: $\delta = +10\%$ (plant $\hat{\mathbf{M}}_i^s$ multiplied by 1.1), controller unchanged. Saturation limits active.

Observations (Fig. 3, $\delta = +10\%$ with saturation):

- Despite the plant being 10% heavier (simulating aging), the CTC controller **still tracks the desired trajectory** — all joints converge to their reference values.
- Tracking errors remain bounded and small; the controller compensates for the parameter mismatch through feedback.
- Torques adjust to handle the heavier plant, remaining within saturation limits.
- This demonstrates that CTC is **robust to moderate parameter mismatch** in practice — the PD feedback term in $\ddot{\theta}^*$ provides sufficient correction even when the model-based feedforward is imperfect.

3 Part (c): PD Control Comparison

Method. PD+gravity compensation: $\tau_{PD} = \mathbf{K}_p^{PD} \mathbf{e} + \mathbf{K}_v^{PD} \dot{\mathbf{e}} + \mathbf{N}_{grav}(\boldsymbol{\theta})$, with $\mathbf{K}_p^{PD} = 5000\mathbf{I}$, $\mathbf{K}_v^{PD} = 200\mathbf{I}$. Unlike CT, PD does not use $\mathbf{M}$ or $\mathbf{C}$ for decoupling. Both controllers are compared **without saturation** to isolate control strategy differences.

Results.

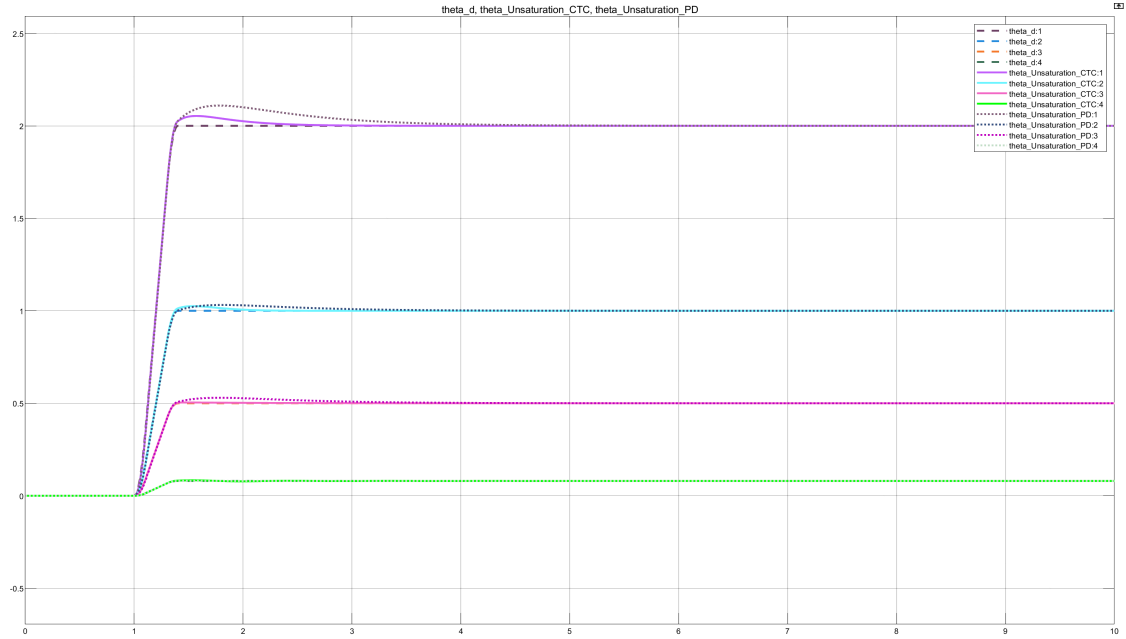


Figure 4: CTC vs. PD comparison (no saturation). **Dashed:** θ_d (reference). **Solid:** CTC. **Dash-dot:** PD.

Observations:

- **CTC:** tracks the reference almost perfectly — smaller overshoot, faster response, and quicker convergence thanks to exact dynamic cancellation.
- **PD:** also converges to the reference, but with noticeably larger transient deviation and slower settling due to uncompensated Coriolis/centrifugal coupling.
- Both achieve zero steady-state error (PD benefits from gravity compensation).
- **Overall: CTC outperforms PD** in tracking accuracy, transient smoothness, and convergence speed, at the cost of higher computational complexity and model dependency.

4 Part (d): Torque Saturation

Method. Saturation blocks limit $|\tau| \leq [6000, 2000, 1000, 100]^\top$ in all simulations. This is active throughout Parts (a)–(c).

Results. The effect of saturation is evident in the preceding sections:

- **Fig. 1:** θ (saturated) deviates from θ_{unsat} (unsaturated) during the step transient — most visible on joint 2, which exhibits a temporary dip when saturation clips the torque.
- **Fig. 2:** the torque limits $[6000, 2000, 1000, 100]^\top$ are reached during the transient, confirming that saturation is active.
- **Fig. 3:** under $\delta = +10\%$ plant perturbation with saturation active, tracking degrades from the combined effect of model mismatch and torque clipping.

Finding:

- Saturation is **not negligible** — torque limits are hit during the step transient, causing visible tracking deviation, especially on joint 2.
- Joint 1 (base) has the highest torque demand; joint 2 is most visibly affected by clipping.
- Saturation slows the response and introduces transient tracking errors that would not exist under ideal unbounded actuation.

5 Conclusion

Computed torque control achieves near-perfect tracking when the model is exact, with smaller overshoot, faster response, and quicker convergence than PD. Even with 10% plant parameter mismatch (simulating aging), CTC continues to track effectively — the PD feedback term provides sufficient correction. PD control trades peak accuracy for robustness and simplicity. Torque saturation is active during the step transient under the given limits $[6000, 2000, 1000, 100]^\top$, causing visible tracking deviations (notably on joint 2); tighter limits would degrade performance further. Future work: adaptive/robust CT extensions and anti-windup schemes.